Te Tāhuhu o te Mātauranga
Ministry of Education

The New Zealand Curriculum

Technology Years 0–10

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Te Kāwanatanga o Aotearoa

About this document

This document provides the Years 0–10 Technology learning area of The New Zealand Curriculum. Tau 0–10 Hangarau wāhanga ako for Te Marautanga o Aotearoa is being published separately.

The content of this document is proposed to be made by the Minister of Education as a National Curriculum Statement under section 90(1)(a) of the Education and Training Act 2020. Once made and published in the New Zealand Gazette, it will set the requirements for the teaching of Years 0–10 Technology in state schools, and schools will be required to start using it in Years 9–10 from 1 January 2027 and in Years 0–8 from 1 January 2029.

Depending on their contexts, some schools may not be ready for full implementation on day one of the specified school year, so the following transition arrangements are proposed to apply:

- For students in Years 0–8, schools will need to make a start by developing, and then executing, a plan that achieves full implementation as soon as practicable and by no later than the end of 2030.
- For students in Years 9–10, schools will need to make a start by developing, and then executing, a plan that achieves full implementation as soon as practicable and by no later than the end of 2027.

Implementation requirements for specified kura boards will be confirmed when the proposed National Curriculum Statements are made under the Act and published in the New Zealand Gazette. This is to allow more time for the Crown to engage with Kaupapa Māori education providers through their representative bodies in advance of these proposed decisions affecting their kura. A specified kura board means the board of any of the following:

- a kura kaupapa Māori
- a designated character school with a character that is hapū- or iwi-based or that affiliates with Ngā Kura ā-Iwi o Aotearoa
- a state integrated school with a special character that is hapū or iwi-based.

This document will be replaced with the official version made under the Act and published in the New Zealand Gazette once that has occurred.

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Purpose statement

Tukuna a whakaaro auaha kia rere kia whakaumutia ai te ao.
Through creativity and innovation, we intervene to transform the world.

Technology is the study of how people use knowledge, resources, systems, and processes to respond to needs, opportunities, and challenges. Students learn how technological outcomes are designed, developed, evaluated, and improved, and how they shape, and are shaped by, society and the environment.

Technology fosters curiosity about how things work, how they are made, and how they can be improved. It develops creativity, innovation, precision, and judgement in the design and development of technological outcomes. It empowers students to apply knowledge, experiment with ideas, evaluate evidence, and see themselves as capable contributors to the future.

Technology equips students with the knowledge, practices, and dispositions needed to understand and influence their designed world as emerging technologists and innovators. As students progress through Technology, they deepen their understanding of how innovation and technological outcomes are enhanced through interactions between people and cultures. Students learn how people identify opportunities, respond to challenges, and create technological outcomes that influence the lives of individuals, communities, and society.

The Technology learning area prepares students with the knowledge and practices to access related curriculum subjects for Years 11–13.

Learning Area Structure

Strands

In Years 0–6, the teaching sequence is organised as one strand.

- **Design, Make, and Innovate:** Introduces foundational knowledge about design, materials, ingredients, and technological outcomes. Students learn how ideas are explored, tested, and refined to meet needs, and how design choices affect people, places, and environments.

In Years 7–10, the teaching sequence is organised into six strands. This includes Design, Make, and Innovate as an underpinning strand and five discipline-specific strands.

- **Design, Make, and Innovate:** Provides the common body of technological knowledge and practice that underpins all other strands. Students develop knowledge of the relationship between people, place, and technology, understanding how technological interventions both shape – and are shaped by – their contexts. They consider ethical and responsible decision-making by asking, 'We can do this, but should we?' The strand develops understanding of how ideas are generated, modelled, tested, evaluated, and refined using specifications, constraints, evidence, and stakeholder input to produce fit-for-purpose outcomes.
- **Materials Technology (Resistant and Textiles):** Focuses on the safe and purposeful selection, processing, and use of materials in designing and developing outcomes. It develops understanding of material properties, construction, and joining techniques, tools, and processes, and how these influence the function, appearance, durability, quality, and sustainability of outcomes.
- **Food and Processing Technology:** Focuses on designing and developing processed food outcomes through responsible and sustainable practices. It develops understanding of ingredients, techniques, and controlled conditions to meet functional and sensory requirements.
- **Systems and Control:** Focuses on how mechanical, electronic, and digital systems operate as interacting subsystems through inputs, processes, outputs, control, and feedback. It develops understanding of forces, movement, circuits, and system behaviour, applied through building, testing, and refining purposeful systems that are safe and reliable.
- **Digital Technology:** Focuses on designing and developing digital outcomes using digital media, digital systems, data, algorithms, and programming. It develops understanding of how data is represented and processed, and how digital components work together to solve problems. Students develop knowledge of computational thinking, algorithms, data representation, testing, and evaluation, and apply this knowledge through responsible digital practices.
- **Spatial and Product Design (Year 9–10 only):** Focuses on form, function, and aesthetics in representing and refining design ideas. It develops understanding of drawing and modelling to communicate and develop outcomes in response to user needs and contexts.

Each student in both Years 9 and 10 must be taught the **Design, Make, and Innovate** strand through at least two of the discipline-specific strands as set out in this learning area. The discipline-specific strands can be different in Year 9 and Year 10.

Organisation of the teaching sequence

The teaching sequence is organised by two components – knowledge and practices.

Knowledge sets out the scope of content to be taught. These statements are more expansive than in past curricula, because they are specific about how deep to go at each year level. The practices set out the disciplinary skills and applications to be taught. Neither the knowledge, nor the practices, are written as achievement objectives. They should be read as teaching requirements.

The knowledge and practices relate to each other but are not an exact match. Sometimes a practice will connect to more than one knowledge statement. Other times there will be knowledge statements that do not have an aligned practice. This is because much of the knowledge will be developed and consolidated through explicit teaching or the general learning activities related to this teaching. The practices underpin the ways of working used by technologists and innovators in related disciplinary fields. They reflect how technological professionals investigate opportunities, generate ideas, develop solutions, test outcomes, evaluate impacts, and refine innovations over time.

The direction of the relationship between knowledge and practices is not fixed. Sometimes a teacher will begin by teaching the knowledge then use the practices to consolidate it. Other times the teacher will start by teaching a practice and making explicit the knowledge. It is the teacher's decision-making that establishes the direction of travel between knowledge and practice, makes knowledge explicit, and brings the teaching sequence to life.

The importance of community context

Teachers enrich the teaching sequence and give it meaning by connecting it to directly relevant and visible examples from the local context: people and place, te ao Māori, and experiences in the lives of students and their communities.

It is important to note that while the teaching sequence includes prescribed knowledge and practices, examples and contexts that include an 'e.g.' are illustrative. These examples are intended to provide guidance but not to mandate what is to be taught. Provided the intended knowledge and practices are taught, teachers may substitute alternative examples that better reflect local environments, community, hapū and iwi connections, student interests, teacher expertise, or emerging issues relating to the discipline.

Glossary

To support consistency in understanding disciplinary terminology, please refer to the supplemental glossary provided.

Technologists

Technology explores how people use knowledge, resources, systems, materials, and processes to develop technological outcomes that meet needs, create opportunities, and address challenges. Technologists and innovators draw on disciplinary knowledge, creativity, evidence, and systems thinking to design, develop, evaluate, and refine outcomes that shape, and are shaped by, people, places, and environments.

Across Phases 1–4, students develop knowledge of how technologists and innovators respond to needs, opportunities, and challenges. They learn how technological outcomes are designed, developed, evaluated, and refined, and how these outcomes affect people, places, and environments. As their understanding grows, students increasingly apply this knowledge through technological practices such as generating and evaluating ideas, using evidence to inform decisions, collaborating with others, and developing technological outcomes.

Note on digital devices

The Technology learning area has been developed so as not to require the use of digital devices in Years 0–5. Teaching in Computational Thinking does provide foundational knowledge and practices for later study in Digital Technology but has been written to be taught with or without student access to digital devices. From Year 6 onwards, digital devices will be required to access teaching and learning in certain strands.

Overview of each phase of teaching

In Phase 1 (Years 0–3), teachers support students to: notice how everyday objects are designed, made, and used, and to understand why they matter. Through guided designing and making experiences, students develop knowledge of materials, tools, simple processes, and the purposes technological outcomes serve. Through observation, experimentation, and purposeful making, they begin to understand how ideas can be turned into technological outcomes for real purposes. Students start to see themselves as creators and problem-solvers who can make decisions about what, how, and why things are made.

In Phase 2 (Years 4–6), teachers support students to: move from curiosity to increasingly purposeful creation. Students develop knowledge of audiences, contexts, constraints, consequences, and technological impacts when responding to identified needs and opportunities. They increasingly act like young technologists, considering trade-offs and judging and evaluating how design decisions affect people and environments. Students test, adapt, and refine designs over time, deepening their understanding of how technological outcomes are shaped by purpose, impact, and the needs of others.

In Phase 3 (Years 7–8), teachers support students to: engage in intentional technological practice. Students develop knowledge of design briefs, specifications, constraints, and frameworks, and use these to generate, model, test, and refine outcomes. They strengthen their identity as emerging technologists and innovators by evaluating impacts and responding to authentic challenges. Students engage with different perspectives and contexts, consider consequences, and explore the question, ‘We can do this, but should we?’ Through this, they learn to balance function, aesthetics, and impacts when making informed design decisions.

In Phase 4 (Years 9–10), teachers support students to: apply technological knowledge and practices in increasingly complex authentic contexts. Drawing on New Zealand contexts, including te ao Māori and Māori innovation, students develop deeper understanding of how cultural perspectives, technological systems, and emerging technologies shape design and expand human possibilities. They investigate ideas through research, modelling, testing, and refinement. Students evaluate the impacts of technological change and weigh trade-offs between ethical, sustainable, functional, cultural, and user-centred considerations. They increasingly recognise the roles of technologists, innovators, designers, and makers in shaping the future. Through this, they take greater ownership of design decisions and develop fit-for-purpose outcomes that are responsive to people, place, and context.

Capabilities across the phases

Across the four phases, Technology provides rich opportunities for students to strengthen the capabilities that support understanding, innovation, and informed action in a changing world. Through designing, developing, evaluating, and critiquing technological outcomes and systems, students develop **communication** as they justify ideas and decisions, **collaboration** as they work with others to address challenges, **self-management** as they plan, test, and refine solutions, **problem-solving** as they respond to authentic technological needs and constraints, and **creativity** as they generate, improve, and apply ideas. Together, these capabilities enable students to contribute confidently to technological innovation and society.

Phase 1 (Years 0–3) Teaching sequence

Design, Make, and Innovate

	Knowledge *The facts, concepts, principles, and theories to teach*			Practices *The skills, strategies, and applications to teach*		
	During Year 1	**During Year 2**	**During Year 3**	**During Year 1**	**During Year 2**	**During Year 3**
	Design • Technology is created by people to meet needs or solve everyday problems as seen in everyday school and community solutions such as shelters designed to protect from rain or playground equipment designed to be safe. • All made objects begin with a design thought, idea, or experience. • Designs are made through a series of steps that are done in a particular order (a design process). • Following steps helps plan, draw, and make a design. **Materials and ingredients** • Products are made of different materials or ingredients for different purposes (e.g. a shoe, a bag, a muesli bar). • Materials are used for different purposes (e.g. durable materials to not break, bendy and stretchy materials for flexibility) (see Year 1 Science: Objects). • Ingredients can be combined in different ways for different products (e.g. carrots in a soup or carrots in a cake).	**Design** • Designs are influenced by the needs of users and the situation, as seen in toys and games that were developed using materials available at the time and designed for play, learning, and social interaction (e.g. bone – knucklebones, wood – tī rākau (stick game), fabric – kites) (see Year 2 Social Sciences: New Zealand Histories). • Products can be described by how they look and how they work. • Different drawings can show different aspects of a design (e.g. how it looks or how it works). • A design brief explains what is being made, who and where it is for, when it is being made, and why. **Materials and ingredients** • Products work best when a specific combination of materials or ingredients is chosen. • Food outcomes are made of different ingredients for different purposes (e.g. spices are added to change the flavour of a product). • Materials or ingredients can be changed using different techniques and processes to make something new (e.g. cutting a piece of cardboard, mixing flour, eggs, and milk). • Different cultures may have different protocols around the cultivation, gathering, and use of materials and ingredients. Within te ao Māori, this is seen in tikanga, such as harvesting only the external fronds of harakeke.	**Design** • Technological outcomes are designed solutions created for a purpose, as seen in early Māori innovation in food-gathering technology (e.g. hīnaki (woven eel trap), matau (fishhook), toki (adze), kupenga (fishing net)), which used local materials (e.g. mussel shell, muka (processed flax fibre)) and responded to natural environments (e.g. shapes of rivers) (see Year 3 Social Sciences: New Zealand Histories). • Technological outcomes may be products, systems, environments, services, or combinations of these. • Technology can have positive and negative impacts on people and the environment. • Design ideas are often shown visually using drawings, models, or photographs to explain what is being made. • Whether something is well made and fit for purpose can be determined by how it looks (aesthetics) and how well it works (function). **Materials and ingredients** • Materials have different performance properties (e.g. strength, flexibility, conductivity) which affect how they are used in material outcomes (see Year 3 Science: Material Properties). • Ingredients have different performance properties (e.g. texture, flavour, moisture) which affect how they are used in food outcomes. • Some changes to materials or ingredients can be undone (e.g. melting chocolate), but others cannot (e.g. cutting a piece of paper). **Computational thinking** • Clear, step-by-step instructions are used to complete tasks consistently. • When instructions do not work as expected, bugs can be found and corrected (debugged).	**Design** • Talk about what a product is used for and who it helps. • Use prompting questions (e.g. 'the five Ws': what, where, why, when, who) to consider how design ideas solve everyday problems. • Create ideas through drawing, playing, tinkering, modelling, and role playing to explore opportunities and communicate ideas. • Draw ideas to help plan what to make. • Follow steps to plan, make, and improve a design. • Describe a completed product and what it does. **Materials, ingredients, and making** • Investigate a range of materials through sorting, shaping, and testing. • Safely make a simple product based on an idea.	**Design** • Identify who a product is for and explain how it helps them in a given context. • Explore everyday products to identify who and what they are for and how they work (e.g. school bag, scissors). • Draw a picture (design) of a product that shows how it might look and work. • Follow the steps of a design brief to make a product. • Describe a completed outcome, who benefits from it, and identify any challenges when using it. **Materials, ingredients, and making** • Investigate how materials can be cut, shaped, or joined when making a product. • Investigate how ingredients change when they are mixed or heated (e.g. flour, milk, eggs). • Safely make a product using a combination of either materials or ingredients. • Explore appropriate ways to use materials and ingredients, including observing tikanga where appropriate.	**Design** • Identify how a proposed outcome might help people and any problems it could cause. • Investigate a range of outcomes for their impact on people and the environment (e.g. considering lunchbox rubbish from pre-packaged foods vs ease of preparing lunch). • Compare outcomes with the same function from different time periods to see how they have changed and who uses them. • Communicate design ideas by drawing, building models, or making simple plans that show different aspects of a design. • Identify what worked or did not in a completed outcome and explain how it could be improved. **Materials, ingredients, and making** • Select materials or ingredients for making a specific outcome based on their performance properties. • Observe and explain how ingredients change when mixed or heated and whether they can change back or not. • Safely make a fit-for-purpose outcome, using either materials or ingredients. **Computational thinking** • Break a simple task into ordered steps, giving clear instructions, and checking and correcting them as they are followed. • Identify where there is a bug and follow logical steps to remove it.

Phase 2 (Years 4–6) Teaching sequence

Design, Make, and Innovate

	Knowledge *The facts, concepts, principles, and theories to teach*			Practices *The skills, strategies, and applications to teach*		
	During Year 4	**During Year 5**	**During Year 6**	**During Year 4**	**During Year 5**	**During Year 6**
	Design • Technological design decisions affect people, places, and environments, as seen in reusable or recyclable packaging products designed to reduce single-use packaging waste in schools and communities. • Technologists and innovators continually develop new technologies (e.g. prosthetic devices, text-to-speech technologies) in response to changing needs and opportunities. • Purposeful design balances user needs with impacts on people, places, and the environment. • A design brief can be used to test ideas and get feedback from others before making the final outcome. • Sketches and models can be used to show different design ideas and explore possibilities. **Materials and ingredients** • Materials and ingredients are used for different purposes depending on their performance properties (e.g. strength, flexibility, conductivity). • Raw materials and ingredients come from different origins (e.g. wool comes from animals such as sheep, flour is made from grains such as wheat). **Systems thinking** • Simple systems (e.g. a toaster) are made of interacting parts that work predictably and consistently to produce an output. **Computational thinking** • Computational thinking problems can be solved using instructions (algorithms) that include ordered steps (sequence). • Complex problems can be broken down into smaller, more manageable parts (decomposition).	**Design** • Technology develops over time as ideas are modified to better meet user needs and contexts, as seen in the ongoing improvement of transport, farming, and outdoor recreation technologies. • People create outcomes for a range of reasons (e.g. to help solve problems, to bring original design ideas to life). • A design can begin with lots of ideas (divergent thinking) and then testing and improving them (refinement). • Sketches, labelled diagrams, simple plans, and models communicate different ideas about a design. • Timelines and flowcharts can be used to plan, communicate, and review design and making processes. **Materials and ingredients** • Different materials and ingredients are processed or manufactured in different ways (e.g. muka is extracted from harakeke, tofu produced from soybeans), which changes their properties and affects how they perform in an outcome. • Processing methods and conditions (e.g. time, temperature) influence how ingredients change and perform in an outcome. **Systems thinking** • A simple system has controls that vary how it works (e.g. dial on a toaster). • Structural systems use parts like beams and columns to stay strong and hold weight (e.g. a bridge is made up of different structures and shapes that create the strength and flexibility). **Computational thinking** • Challenging computational thinking problems can be solved using instructions (algorithms) that include ordered steps (sequence), repeated actions (loops), and decisions (selection). • Challenging computational thinking problems are decomposed to solve them more effectively.	**Design** • Technological outcomes reflect values and priorities, involving trade-offs between factors like performance, cost, community well-being, and sustainability, as seen in different approaches to food production (e.g. hua parakore (kaupapa Māori organic farming), hydroponic horticulture, pastoral farming, glasshouse production). • A design brief can be used to evaluate what worked well and what could be done better next time. • Designs can be evaluated against identified attributes, including aesthetics, function, and fitness for purpose, which can be developed into specifications. • Different types of visual design representations (e.g. models, sketches, diagrams) are used to communicate information about a design at different stages of development. **Materials and ingredients** • Materials and ingredients are selected based on their performance, production methods, and impacts on people and the environment. • Materials can fail or wear over time depending on how they are used, joined, maintained, repaired, or reused. **Systems thinking** • Some components operate as a 'black box', a hidden system where inputs and outputs are known, but the internal functions are not (e.g. a remote control). • Mechanical and electronic systems use components to transfer energy, control movement, and produce outputs. **Computational thinking** • Digital systems and programs use sequence, loops, and selection to process information, make decisions and perform tasks. • Computational thinking includes using abstraction to focus on the important information needed to solve a problem. **Digital technology** • Digital systems use software and applications to process data, perform tasks, and meet user needs. • All digital media is affected by copyright laws. In New Zealand, copyright protects original work as soon as it is recorded, without registration.	**Design** • Analyse how a design choice may have positive and negative impacts. • Collect feedback on a design idea and use that feedback to improve the design. • Write a design brief for a need or opportunity, with description of physical and functional attributes. • Use sketches or models to test a design idea and give reasons for choosing which to develop. **Materials, ingredients, and making** • Test materials and ingredients for particular performance properties. • Make an outcome using materials or ingredients that draws on feedback from others. **Systems thinking** • Investigate the inputs, controls, and outputs of a system (e.g. door handle, simple light circuit), explaining how they work together. **Computational thinking** • Recognise repeating patterns in familiar visual designs (e.g. tukutuku panels, tapa cloth, sashiko stitching) and describe how they may be drawn through ordered steps. • Develop algorithms that use sequence to solve computational thinking problems. • Decompose complex problems to better understand how to approach them.	**Design** • Compare how an outcome has changed over time and explain why those changes were made and whether they improved fitness for purpose. • Write broad design criteria (attributes) to guide and refine a design idea. • Use a timeline or flowchart to guide, record, and review design decisions throughout a project. • Sketch and discuss ideas to make changes to a design for a reason. • Use drawings and models in different ways to explore ideas about possible design, identify problems, and suggest changes. **Materials, ingredients, and making** • Test materials to compare physical properties (e.g. strength, flexibility, durability) and use findings to justify material selection for an identified purpose. • Compare different materials or ingredients to explore their impacts on environment (e.g. whether they are reusable or compostable). • Compare how different processing methods or conditions affect the outcome of a product. **Systems thinking** • Identify the component features of a simple system and predict how they impact the way the system works. • Draw and label diagrams to show components and connections within a simple system. • Design and test a simple structural system (e.g. a tower or a bridge to hold a weight). **Computational thinking** • Develop algorithms that use sequence, loops, and selection to solve complex computational thinking problems. • Decompose challenging computational thinking problems to understand how to approach them effectively.	**Design** • Justify design decisions using feedback and considering user needs, impacts, and trade-offs. • Analyse the wider implications of using new technologies (e.g. social, environmental, ethical). • Develop a design brief to explain what is being made, who it is for, when and where it is to be used, and why it meets user needs. • Use a range of visual design representations and simple models to communicate different aspects of a design idea. • Evaluate an outcome against its identified aesthetics and function and judge whether it is fit for purpose. **Materials, ingredients, and making** • Select and justify materials, considering performance requirements and impacts on people and environment (e.g. recycled materials). • Explain how processing methods and conditions influenced the attributes of an outcome and suggest improvements. • Make and evaluate an outcome against its intended attributes, judging whether selected materials were fit for purpose. **Systems thinking** • Make a simple outcome with moving parts. • Construct and test a simple electronic system using a kit, identifying its inputs, outputs, and controls. **Computational thinking** • Apply debugging techniques to find and correct errors and improve how an algorithm performs. • Use abstraction to simplify a computational thinking problem. **Digital technology** • Consider who created a digital media text or tool, evaluating whether it can be used for a specific purpose. • Justify the selection of digital tools used for a digital media outcome, in relation to intended audience and purpose. • Select and use digital information from appropriate sources, recording where it came from, and why it can be trusted.

Phase 3 (Years 7–8) Teaching sequence

Design, Make, and Innovate

	Knowledge *The facts, concepts, principles, and theories to teach*		Practices *The skills, strategies, and applications to teach*	
	During Year 7	**During Year 8**	**During Year 7**	**During Year 8**
	The Design, Make, and Innovate strand underpins all other strands in Technology. It provides a body of knowledge and practice that is common to all disciplines. The content in this strand is intended to be put in action through authentic technological practice including designing, making, and developing authentic outcomes within the disciplinary strands of Systems and Control, Materials Technology, Food and Processing Technology, and Digital Technology. Technological outcomes may be products, systems, or a combination of both, developed for a purpose within authentic contexts. Safe and purposeful technological practice is embedded across all strands, requiring learners to identify and manage risks while using materials, tools, equipment, and workspaces appropriately throughout the design and development process. Alongside safe and purposeful practice, learners are also expected to consider ethical, cultural, and social responsibilities in technology development (e.g. concepts such as mana raraunga (Māori data sovereignty), which recognises raraunga Māori (Māori data) as taonga and seeks to affirm Māori rights to govern and protect that data).			
	Innovation in technology • Technological innovation creates outcomes that have intended and unintended impacts on people, places, and environments, as seen in social media platforms designed to connect people, which have also led to unintended effects (e.g. compulsive use, distraction, and other impacts on well-being). • Technological interventions affect people, places, and environments differently. The impacts can be positive or negative and intended or unintended. • Professional bodies set codes requiring technologists to identify and reduce foreseeable harm. Technologists vary in how closely they follow those codes, and some harms are not foreseen. • Innovation happens both in teams and alone. Teams can bring more diverse knowledge and perspectives to a problem and can take longer to settle on a solution. • Artificial intelligence (AI) functionality is embedded in everyday digital tools and devices and responds to data and user inputs. **Designing and developing fit-for-purpose outcomes** • Design principles and design frameworks provide a structured process for developing fit-for-purpose outcomes. • Different drawing types carry different information, such as sketches showing form and proportion, dimensioned drawings carry measurements, and nets communicating how 2D shapes transform into 3D forms. • Fitness for purpose is judged against broad attributes that describe how well an outcome meets user needs and purpose (e.g. reliability, efficiency, usability, safety, cost-effectiveness, sustainability). • Measurable specifications build on identified attributes by providing valid criteria for judging how outcomes are fit for purpose.	**Innovation in technology** • People–place–technology relationships describe how users, cultural perspectives, histories, contexts, and environments shape design decisions, influence priorities, and influence what makes an outcome fit for purpose. This is seen in projects that use virtual reality technologies to create digital representations of physical spaces (e.g. digital museums allowing digital tourism, digital marae enabling people to connect to their whakapapa). • The priorities, values, impacts, opportunities, and constraints relevant to a design depend on its purpose and context and may change as the design develops. Designers make and justify decisions about how these considerations should shape their work. • AI technologies can produce different outputs depending on the data and instructions they are given and may create benefits, risks, and unintended consequences. **Designing and developing fit-for-purpose outcomes** • Design frameworks and design principles consider purpose, needs and opportunities, feedback, and constraints. • Specifications are specific and measurable criteria (e.g. red or blue, weighs 1.2kg, over 75 degrees, 120mm length, travels 30mm) that enable reliable testing and evaluation. • Visual modes (e.g. sketches, drawings, diagrams, simple models) are used to generate, explore, represent, and communicate 3D design ideas from different viewpoints. • Ergonomics and usability principles (i.e. are used to evaluate comfort, safety, ease of use, and accessibility). Anthropometric and usability data can provide evidence for decisions about fitness for purpose. • Functional modelling supports testing, exploration, refinement, and informed decision-making (e.g. accuracy, efficiency, style, building layout, and waste reduction) before resources are committed to final production.	**Innovation in technology** • Evaluate an existing technological outcome to identify its impacts and judge its fitness for purpose. • Evaluate an existing outcome to identify positive and negative, intended and unintended impacts and explain how different users may experience those impacts (e.g. how a plastic container might be used in different cultures). • Investigate how AI is embedded in everyday technologies and describe how it may affect users. **Designing and developing fit-for-purpose outcomes** • Use design frameworks with guidance to analyse existing solutions, considering context and culture, and to guide iterative development with potential users (e.g. graphic organisers for construction plan, food order, flow chart, Positive Minus Interesting). • Generate and explore design ideas using quick sketches and drawings in response to a design context. • Refine or adapt a brief in response to user needs and context. • Develop simple specifications from identified attributes. • Evaluate outcomes against broad attributes, justifying material and technique choices.	**Innovation in technology** • Engage with user needs, cultural perspectives, histories, contexts, and environments to compare priorities and inform innovative design decisions. • Identify and weigh the considerations relevant to their evolving design, justify the priorities and trade-offs they make, and revise their decisions as new possibilities, evidence, and perspectives emerge. • Consider how AI tools may affect people, communities, and environments when deciding how and when to use them. **Designing and developing fit-for-purpose outcomes** • Use a design framework and design principles to make and justify decisions about function, aesthetics, feasibility, sustainability, impact, and user needs. • Apply visual modes to represent designs more accurately and communicate the form and function. • Use specifications to guide the development of a fit-for-purpose outcome (e.g. aesthetic, functional, ergonomic, safety requirements). • Use functional modelling (e.g. cardboard or play dough models, sketching) to test design intent, accuracy, and material efficiency before final production. • Test and evaluate against the specifications to identify the potential feasibility of the outcome (e.g. misalignments or unintended consequences, improving accuracy, fairness, reliability).

Materials Technology (Resistant and Textiles)

	Knowledge *The facts, concepts, principles, and theories to teach*		Practices *The skills, strategies, and applications to teach*	
	During Year 7	**During Year 8**	**During Year 7**	**During Year 8**
	The content within this strand is underpinned by the common body of knowledge and practice in the Design, Make, and Innovate strand. These foundational design and development processes are to be applied here in the context of creating authentic and fit-for-purpose Materials outcomes. Safe and purposeful practice includes identifying and managing risks, and applying best-practice use of materials, tools, equipment, and workspaces throughout the designing, developing, constructing, and evaluating of outcomes.			
	• Performance properties of materials (e.g. strength, flexibility, absorbency, conductivity, heat stability) affect how well an outcome works and is able to be maintained. • Designers consider user needs, user values (e.g. quality), and environmental impacts when selecting materials (e.g. lifecycle, durability, waste, repair, reuse, disposal). • Sustainable material choice weighs fitness for purpose against environmental impact across a material's whole life. Recycled, recyclable, biodegradable and renewable options often reduce impact but not always, and lifecycle evidence is needed to tell which. • Different construction and joining techniques (e.g. stitching, gluing, folding, interlocking) are used for different materials and purposes, influencing how parts of an outcome are assembled and held together. • Material properties and construction techniques work together to influence the function, appearance, durability, and quality of an outcome (see Year 7 Science: Material Properties and Thermal Energy).	• The composition, structure, and chemical treatment of materials determine their performance properties. • Material properties and attributes influence the form, function, quality, and sustainability of a designed outcome, including how suitable it is for its intended user and environment. • Functional modelling is used to determine the techniques and processes required to meet specifications (e.g. accurate measuring, cutting, shaping, joining, forming, finishing, quality checking). • Selection of different materials and construction techniques is informed by aesthetic and functional requirements of the outcome (e.g. performance requirements, user needs, environmental impact, lifecycle, recycled or recyclable materials). • Testing and analysis of material or outcome failure informs decisions (e.g. refine, redesign, modify, adapt) by identifying weaknesses (e.g. in materials, construction, joining methods, safety, hygiene, fitness for purpose).	• Design and make an authentic, fit-for-purpose Materials Technology outcome. • Use knowledge of different materials (e.g. composition, structure, properties) to choose suitable joining, construction, shaping, or forming methods. • Investigate how form and function can be defined by the properties of different materials. • Balance user values and environmental impacts with functionality when making design decisions. • Explore sustainable material choices (e.g. recycled, recyclable, biodegradable, renewable, low-waste materials) to determine fitness for purpose. • Investigate how different construction and joining techniques influence the form, function, and durability of an outcome.	• Design and make an authentic, fit-for-purpose Materials Technology outcome. • Test the performance properties of materials with similarities and differences in composition, structure, and chemical treatments. • Select and justify material and construction choices by explaining how properties and attributes influence form, function, quality, longevity, and user suitability. • Use functional modelling and construction techniques and processes under guidance to accurately create a fit-for-purpose resistant or textiles materials outcome to meet specifications. • Investigate sustainable material options to consider their use in an outcome and context. • Test materials, joints, construction methods, and outcomes to identify weaknesses, explain failure, and inform redesign for improved safety, usability, durability, or fitness for purpose.

Food and Processing Technology

	Knowledge *The facts, concepts, principles, and theories to teach*		Practices *The skills, strategies, and applications to teach*	
	During Year 7	**During Year 8**	**During Year 7**	**During Year 8**
	The content within this strand is underpinned by the common body of knowledge and practice in the Design, Make, and Innovate strand. These foundational design and development processes are to be applied here in the context of creating authentic and fit-for-purpose Food and Processing Technology outcomes. Safe, purposeful and hygienic practices (e.g. handwashing, hairnet use, knife safety) support the production of processed food outcomes that are safe to consume, including the appropriate use of ingredients, equipment, processing techniques, and food-handling procedures.			
	• Sensory requirements of an outcome (e.g. flavour, aroma, appearance) are important attributes that can be met through careful ingredient selection. • The attributes of existing food outcomes can help to inform decisions when selecting ingredients to produce a fit-for-purpose outcome (e.g. crackers are savoury, Granny Smith apples are tart, lavender is fragrant). • Different ingredients contribute specific performance properties that influence flavour, texture, appearance, and overall success of a processed food outcome. • Processing techniques contribute to the attributes of processed food outcomes and the overall success of the outcome, such as blended to be smooth, grilled to be golden, cultured through fermentation to produce yoghurt, or cooked using hāngī or ‘umu to develop distinctive flavour, texture, and moisture. • Food waste is one of the costs that designers of processed food outcomes weigh and can be minimised when designing new processed food outcomes.	• Sensory and objective measures (e.g. size, weight, temperature) can be used to generate specifications and compare processed food outcomes. • Production processes can be controlled by managing conditions (e.g. time for yeast to ferment, temperature for yoghurt culture, exact temperature to bake carrot cake, or controlling heat retention and cooking time in hāngī or ‘umu) to achieve predictable results. • Ingredients have performance properties that can be tested and compared against specifications to determine fitness for purpose. • Functional modelling trials processing techniques and tests ingredients against broad specifications to determine fitness for purpose before production. • Responsible designers of food outcomes consider user and environmental impacts when selecting ingredients and processing methods, including cultural food technologies.	• Design and make an authentic, fit-for-purpose Food and Processing Technology outcome. • Use sensory analysis of existing food outcomes to identify and describe attributes (e.g. chewy, crunchy, golden brown). • Select ingredients and simple processing methods based on the required attributes for an intended outcome (e.g. grated carrot in a cake for added sweetness). • Compare the performance properties of ingredients in existing food outcomes to inform ingredient selection for an intended outcome. • Investigate processing techniques to determine suitability for the intended outcome to meet the required attributes (e.g. chopping, grating, fermenting, baking, hāngī, or ‘umu cooking methods). • Review ingredient and material choices in existing solutions to suggest responsible improvements (e.g. recyclable, biodegradable, or low-waste ingredients and packaging) while maintaining fitness for purpose.	• Design and make an authentic, fit-for-purpose Food and Processing Technology outcome. • Compare processed food outcomes in relation to required attributes and specifications to determine fitness for purpose and identify possible improvements (e.g. comparing commercial carrot muffins to class-made carrot muffins to determine improvements). • Test and evaluate ingredients against identified specifications to support decisions about refinement and fitness for purpose. • Control processing conditions (e.g. time and temperature, method) to achieve consistent and predicted outcomes. • Use functional modelling to test ingredients and processing techniques against specifications, informing refinements to improve fitness for purpose (e.g. bread fermentation trials, yoghurt cultures, baking trials). • Select ingredients and materials with consideration of the impacts on people and the environment.

Systems and Control

	Knowledge *The facts, concepts, principles, and theories to teach*		Practices *The skills, strategies, and applications to teach*	
	During Year 7	**During Year 8**	**During Year 7**	**During Year 8**
	The content within this strand is underpinned by the common body of knowledge and practice in the Design, Make, and Innovate strand. These foundational design and development processes are to be applied here in the context of creating authentic and fit-for-purpose Systems and Control outcomes. Safe and purposeful practice includes identifying and managing risks with support, following relevant health and safety procedures when using tools, equipment, batteries, lasers, electrical systems, testing devices, and powered components throughout the design, development, construction, and evaluation of outcomes. Knowledge and practices related to programmed components can be taught alongside Digital Technologies, which covers programming. This content can also be taught independently using pre-programmed components.			
	Systems thinking • Systems can be mechanical, electronic, and digital and consist of smaller systems (subsystems) that each perform a specific function and work together to achieve an overall purpose. • Systems and subsystems respond to inputs. A control loop compares a system's output with a target value and adjusts the input to reduce the difference. • Standard symbols represent components and connections at a basic level (see Year 6 Science: Simple Circuits). **Physical systems** • Mechanical motion types (e.g. linear, rotary, reciprocating, oscillating) describe how systems move. • Structural forces (e.g. compression, tension, shear, torsion) describe how systems use, and resist loads (see Year 6 Science: Representing Forces). • Electrical outcomes use electrical energy to perform straightforward functions, while electronic outcomes process and control information through circuits that combine inputs and outputs.	**Systems thinking** • System diagrams explain input–process–output relationships, subsystems, feedback, and control loops. • Digital hardware, software, and networks interoperate to process and share information. • Iterative improvement uses user feedback and performance data to identify possible faults and enhance reliability and usability. **Physical systems** • Component values, colour codes, voltage, current, and resistance support accurate component selection and explain behaviour in electronic systems. • Integrated systems combine mechanisms, structures, circuits, controls, and simple programmable components to transfer energy, create movement, process signals, and respond to inputs.	• Design and make an authentic, fit-for-purpose Systems and Control outcome. **Systems thinking** • Build and test simple systems that incorporate feedback, control, and basic accessibility features. • Represent system functions and connections with standard symbols. • Deconstruct an everyday system to identify interactions between electrical, electronic, mechanical, and structural components. **Physical systems** • Compare how structures support loads, use movement, and resist forces. • Investigate simple machines and mechanisms to show how movement or force changes.	• Design and make an authentic, fit-for-purpose Systems and Control outcome. **Systems thinking** • Produce annotated system diagrams that explain subsystems, reliability risks, and input–output transformations. • Test and refine a system using user feedback or performance data to improve reliability and usability. • Build and test controlled circuits using manual or simple programmable controls. **Physical systems** • Select components using data sheets and colour codes. • Integrate systems (e.g. motor-driven mechanisms) and document mechanisms, movement, forces, control methods, and connections using appropriate conventions.

Digital Technology

	Knowledge *The facts, concepts, principles, and theories to teach*		**Practices** *The skills, strategies, and applications to teach*	
	During Year 7	**During Year 8**	**During Year 7**	**During Year 8**
	The content within this strand is underpinned by the common body of knowledge and practice in the Design, Make, and Innovate strand. These foundational design and development processes are to be applied here in the context of creating authentic and fit-for-purpose Digital Technology outcomes. Safe and purposeful practice includes protecting personal information, managing digital files and data appropriately, using digital systems and online environments responsibly, and following agreed expectations when creating, sharing, testing, and evaluating digital outcomes.			
	• Algorithms can be represented, followed, and tested using plain language, diagrams, block-based or text-based programming languages. • Pattern recognition identifies repeated elements that can be used to simplify problems and develop algorithms. • Digital systems use inputs, processes, outputs, and data to complete tasks and solve problems for users in different contexts. • Digital media outcomes are purposely designed to communicate with specified audiences. • Digital media outcomes follow established design conventions to support user understanding (e.g. layout, readability, usability).	• Different algorithms can solve the same problem. Efficiency is judged by how the number of steps or the memory used grows as the input grows. • Digital systems represent and use data in different forms, including text, numbers, images, and Boolean (true-or-false) values. • Simple programs use variables, input, output, sequence, selection, and iteration to respond to user input or changing conditions. • Unit testing checks that each part of a program behaves correctly on its own. Functional testing checks that the program as a whole meets its requirements. • Improvements to a digital media outcome are made following audience experience testing and evaluation against design conventions.	• Design and make an authentic, fit-for-purpose Digital Technology outcome. • Represent and test simple algorithms using clear sequences, plain language, diagrams, block-based or text-based programming languages. • Identify the patterns found in existing computational thinking activities and evaluate what impact these patterns have on the overall activity. • Examine how the inputs, processes, outputs, and data work together in familiar digital systems. • Create simple digital media outcomes, considering purpose, usability, and accessibility for a specified audience. • Apply design conventions to simple digital media outcomes, considering how files are organised and named, selection of fonts and colour, and basic layout of the elements.	• Design and make an authentic, fit-for-purpose Digital Technology outcome. • Compare algorithms that solve the same problem by counting the steps each takes on inputs of increasing size. Analyse how data is used or represented in a program, digital system, or digital media outcome. • Build and debug simple programs that use variables, user input, output, selection, and iteration to respond to changing conditions. • Test programs by checking that each part of the program and the overall functionality of the program work as intended. • Refine a simple digital media outcome using audience experience feedback and evaluating against design conventions.

Phase 4 (Years 9–10) Teaching sequence

Design, Make, and Innovate

	Knowledge *The facts, concepts, principles, and theories to teach*		Practices *The skills, strategies, and applications to teach*	
	During Year 9	**During Year 10**	**During Year 9**	**During Year 10**
	The Design, Make, and Innovate strand underpins all other strands in Technology. It provides a body of knowledge and practice that is common to all disciplines. The content in this strand is intended to be applied through the design, making, and innovation of authentic outcomes within the disciplinary strands of Systems and Control, Spatial and Product Design, Materials Technology, Food and Processing Technology, and Digital Technology. Safe and purposeful practice is embedded across all strands, requiring learners to identify and manage risks while using materials, tools, equipment, and workspaces appropriately throughout the design and development process. Alongside safe and purposeful practice, learners are also expected to consider ethical, cultural, and social responsibilities in technology development (e.g. concepts such as mana raraunga (Māori data sovereignty), which recognises raraunga Māori (Māori data) as taonga and seeks to affirm Māori rights to govern and protect that data).			
	Innovation in technology • Technological innovation extends human capability, and creates opportunities, risks, and unintended consequences for people, places, and environments. This is seen in the growing use of AI systems across education, business, health, and public service. • Technological change and evolution reshape society by influencing how people live, work, communicate, and interact with the environment. • Emerging technologies develop through design choices, materials, information, data, and processes that affect their performance and impact. • Technological outcomes can be influenced by the quality of information, evidence, testing, and design decisions used during development. • Digital systems, including AI, generate outputs from data, algorithms, models, and user inputs. The quality, accuracy, and reliability of those outputs are influenced by the quality of those inputs. **Designing and developing fit-for-purpose outcomes** • Effective design integrates human and environmental considerations using design frameworks, stakeholder feedback, and evidence to justify decisions. • Designers use design drawings, mock-ups, sketches and modelling to explore, communicate, and refine design possibilities. • Responsible development of fit-for-purpose technological outcomes requires identifying the risks of an outcome, disclosing known limitations, and testing before release.	**Innovation in technology** • Technological systems and innovations are complex, requiring trade-offs between interdisciplinary knowledge, ethical considerations, and long-term societal and environmental consequences, as seen in engineers designing renewable energy systems in New Zealand. • Information quality influences technological decision-making and can affect how technological innovations are developed, adopted, and evaluated. • The adoption of emerging technologies requires evaluation of sustainability, cost, accessibility, ethics, and long-term impacts. • Intellectual property rights influence how technological outcomes are created, shared, used, and protected. • Digital systems, including AI, can create benefits, and risks, including biases and unintended consequences depending on how they are designed, trained, deployed, and used. **Designing and developing fit-for-purpose outcomes** • Design frameworks integrate cultural perspectives, stakeholder feedback, and ethical thinking, to guide evidence-informed design decisions for people and environments. • Designers use relevant 2D and 3D representations (e.g. close-up or viewpoint drawings, models) to communicate, test, justify, and refine design ideas, considering aesthetics, functionality, and intended user experience. • The development of fit-for-purpose technological outcomes requires balancing ethical, legal, environmental, and stakeholder considerations, together with the consequences of malfunction and failure, when making design decisions. • Evaluation of fitness for purpose relies on user information, stakeholder feedback, testing evidence, and relevant data (e.g. anthropometric data (human body measurements) for physical outcomes, user experience (UX) data for digital outcomes).	**Innovation in technology** • Analyse technological innovation to identify intended and unintended impacts on people, places, and environments. • Investigate how design decisions, materials, information, and processes influence the performance and impacts of technological outcomes. • Evaluate the quality of information, evidence, and testing used to support technological development. • Make evidence-informed judgements about opportunities and risks associated with emerging technologies. • Test how data quality, evidence, and user inputs influence the accuracy, reliability, and usefulness of AI-generated outputs. **Designing and developing fit-for-purpose outcomes** • Develop innovative, user-centred outcomes that respond to real-world challenges through collaboration, stakeholder engagement, and iterative improvement. • Apply design thinking, modelling, and structured processes to test, refine, and communicate ideas that meet specifications, and address stakeholder needs. • Use a range of creative modes to generate a variety of design ideas and explore possibilities in response to a design context. • Develop fit-for-purpose outcomes by selecting and integrating tools, materials, ergonomic principles, techniques, and processes.	**Innovation in technology** • Evaluate technological innovations using ethical, cultural, environmental, and stakeholder perspectives to justify design decisions. • Critique the quality and credibility of information used to support technological decisions and evaluations. • Evaluate emerging technologies using criteria such as sustainability, accessibility, ethics, and long-term impacts to justify informed decisions about adoption and use. • Investigate how intellectual property rights influence the development, sharing, and use of technological outcomes. • Evaluate the benefits, risks, biases, and limitations of AI applications to justify informed, ethical, and responsible decisions about their use. **Designing and developing fit-for-purpose outcomes** • Use ethical frameworks and lifecycle analysis to evaluate environmental, social, and cultural impacts and justify design trade-offs. • Use design frameworks and relevant data (e.g. engineering principles, anthropometric data, biotech principles) to justify design decisions and develop fit-for-purpose outcomes that meet specifications. • Apply visual modes to accurately communicate, refine, and justify design ideas considering aesthetics, functionality, and user experience (e.g. scale, materiality, textures, components, colours). • Use testing, malfunction and failure analysis, stakeholder critique, and performance data to justify refinements and evaluate fitness for purpose, demonstrating increasing independence. • Use disciplinary knowledge, stakeholder perspectives, and collaboration with others to inform and justify design decisions.

Spatial and Product Design

	Knowledge *The facts, concepts, principles, and theories to teach*		Practices *The skills, strategies, and applications to teach*	
	During Year 9	**During Year 10**	**During Year 9**	**During Year 10**
	The content within this strand is underpinned by the common body of knowledge and practice in the Design, Make, and Innovate strand. These foundational design and development processes are to be applied here in the context of creating authentic and fit-for-purpose Spatial and Product Design outcomes. Responsible practice includes considering user experience, accessibility, sustainability, cultural perspectives, and the impacts of design decisions when communicating, testing, refining, and evaluating design ideas and outcomes.			
	Spatial and product design • Designers use creative strategies (e.g. forced association, form experimentation) to produce imaginative ideas that differ from earlier solutions reached. • Crating uses simple shapes to establish size and perspective to create well-proportioned 2D and 3D drawings or sketches. Different line weights are used to differentiate construction lines and outlines. • Perspective drawings represent 3D design ideas as seen by the viewer. • Paraline drawings represent 3D design ideas using linear measurements along parallel axes. • Rendered drawings are used to represent the effects of a light source on 3D design ideas. • Existing spatial and product designs demonstrate qualities of form and function that may influence future design decisions.	**Spatial and product design** • Formative divergent thinking involves creative play that is open-ended, to generate new and unique possibilities for potential design ideas beyond first thoughts. • Convergent thinking involves selecting and refining design ideas using design principles relevant to a spatial or product design context. • Orthographic drawings are abstracted representations of 3D design ideas as a set of projected 2D views that are typically arranged in third-angle projection. • 2D and 3D instrumental drawings have fundamental conventions associated with line types and scaling. • Foundational spatial design thinking encompasses the physical attributes of spaces (e.g. circulation, dimensions) and the user's spatial experience (e.g. atmosphere, scale). • Foundational product design thinking encompasses the considerations around the physical attributes of objects and the user's interactions with them. • Designers can be identified by recurring choices of form, material, and proportion across their work.	• Design and make an authentic, fit-for-purpose Spatial and Product Design outcome. **Spatial and product design** • Express and explore creative possibilities through design drawings, mock-ups, or sketches in response to a spatial or product design context. • Generate imaginative ideas using creative strategies (e.g. form experimentation, mixed-media, overlays, visual manipulations). • Use crating and line weights to construct a well-proportioned visual representation of spatial or product design ideas. • Use perspective drawing techniques to represent spatial or product design ideas as seen by the viewer. • Use paraline drawing techniques to represent spatial or product design ideas. • Apply rendering principles to a 3D drawing to convey the effects of a light source on the surface and form of a spatial or product design idea. • Analyse qualities of existing spatial or product designs to inform the generation, evaluation, and refinement of design ideas.	• Design and make an authentic, fit-for-purpose Spatial and Product Design outcome. **Spatial and product design** • Generate new and unique possibilities for potential design ideas using creative play and open-ended exploration. • Select and develop design ideas in relation to the relevant design considerations for a spatial or product design context using design decision-making. • Convey spatial or product design ideas and thinking through sets of relevant 2D and 3D drawings that show visual information through different views and depth of views (e.g. close-ups). • Use orthographic drawing techniques to convey the main outline and dimensions of a 3D design idea. • Evaluate and refine spatial or product design ideas through iterative drawing, modelling, and visual testing. • Apply the techniques of different media to illustrate the visual qualities of a spatial or product design idea. • Analyse the work of an influential spatial or product designer in terms of its characteristic qualities of form and function.

Materials Technology (Resistant and Textiles)

	Knowledge *The facts, concepts, principles, and theories to teach*		Practices *The skills, strategies, and applications to teach*	
	During Year 9	**During Year 10**	**During Year 9**	**During Year 10**
	The content within this strand is underpinned by the common body of knowledge and practice in the Design, Make, and Innovate strand. These foundational design and development processes are to be applied here in the context of creating authentic and fit-for-purpose Materials Technology outcomes. Safe and purposeful practice includes identifying, assessing, and managing risks, and applying responsible and ethical decisions when selecting, processing, constructing, testing, and evaluating materials outcomes, considering impacts on users, communities, and environments.			
	Materials technology • Health and safety codes of practice set out how identified risks are to be controlled. They support industry practice and manage risk assessments. • Material characteristics (e.g. appearance, performance, function) affect user experience and user interaction with an outcome. • Unconventional materials are materials used in new, unusual, or unexpected ways to meet a design need or create an innovative outcome. These can be used alongside more conventional materials in an outcome. • Tools and joining methods are selected against materials and the loads the joins must carry. Different joins (e.g. mechanical fixings, adhesives, thermal joins) each fail in different ways. • Materials with embedded technologies in wearable and non-wearable outcomes influence how data is gathered, used, and shared in different contexts (e.g. tracking devices, biometric data, storing data without permission). • Designers integrate wider values and considerations (e.g. situation, intended user, environment) in the development of material outcomes. • Section shape, join placement, and the distribution of material determine how a material outcome carries load and where it is likely to fail first.	**Materials technology** • Purposeful testing of materials, tools, and techniques builds practical knowledge and helps evaluate feasibility and quality. • Designers make fit-for-purpose choices by understanding material performance properties, codes of practice, and how context can affect how materials perform. • Material selection and lifecycle planning decisions impact longevity, safety, and long-term performance. • Material testing can be used to predict performance, identify failure risks, and justify refinements prior to final production. • Construction methods must align with design intent, material properties, and user needs to ensure quality and durability. • Material choices carry consequences for users, makers, and the environment. Designers weigh these against performance requirements when selecting conventional and unconventional materials.	• Design and make an authentic, fit-for-purpose Materials Technology outcome. **Materials engineering** • Compare material characteristics to identify suitable materials for an outcome and make material selections that will meet specifications (aesthetic and functional). • Choose tools and joining techniques that suit the materials and loads the join must carry and test their effectiveness during construction. • Investigate how materials with embedded technologies gather and use data and respond in different environments and contexts of data collection. • Test materials, tools and joining methods to determine their suitability for an intended outcome. • Measure and mark accurately, select materials based on their characteristics and performance properties, and consider environmental impacts. • Justify material selections by considering performance requirements, user needs, and environmental impacts.	• Design and make an authentic, fit-for-purpose Materials Technology outcome. **Materials engineering** • Use feedback to refine designs, assess how fit for purpose an outcome is, and make responsible decisions about materials and processes. • Select construction techniques and processes to enable an outcome to function as intended, using age-appropriate tools and equipment to construct outcomes with increasing accuracy. • Use functional modelling (including sketching), testing (materials), and trialling (construction techniques and processes) to develop and iteratively refine an outcome's design. • Combine knowledge of material performance properties, sustainability considerations, and techniques and processes to create reliable, fit-for-purpose outcomes. • Analyse material testing and construction evidence to refine outcomes and evaluate fitness for purpose. • Evaluate and apply construction techniques, tools, and methods to refine and optimise outcomes that meet design briefs. • Investigate conventional and unconventional materials to evaluate their suitability for ethical use in a fit-for-purpose outcome.

Food and Processing Technology

	Knowledge *The facts, concepts, principles, and theories to teach*		Practices *The skills, strategies, and applications to teach*	
	During Year 9	**During Year 10**	**During Year 9**	**During Year 10**
	The content within this strand is underpinned by the common body of knowledge and practice in the Design, Make, and Innovate strand. These foundational design and development processes are to be applied here in the context of creating authentic and fit-for-purpose Food and Processing Technology outcomes. Safe, purposeful, and hygienic practices support the production of processed food outcomes that are safe to consume. Responsible practice includes considering cultural, environmental, ethical, and sustainability factors when designing, producing, packaging, storing, and evaluating food outcomes.			
	Codes of practice • Codes of practice guide the safe and appropriate use of equipment to ensure food safety, quality, and reliability. **Ingredients and techniques** • Sensory profiling of existing solutions in relation to required attributes of a food outcome informs ingredient selection (e.g. carrot for muffins, sugar for cookies) for fit-for-purpose outcomes. • Ingredients are manipulated, transformed, or formed by a range of processing techniques (e.g. fermentation, dehydration, emulsification) in the development of a fit-for-purpose outcome. • Ingredients have key performance properties that work together in the development of processed outcomes (e.g. brown sugar for sweet, chewy cookies with a caramel colour). **Responsible design** • Food outcome design decisions influence product quality, resource use, waste, and overall fitness for purpose. • Principles guiding the development of food outcomes (e.g. cultural responsibilities such as tikanga, food safety concerns such as allergens, religious observances such as halal) affect all stages of development, including processing, packaging, storage, and transportation.	**Codes of practice** • Processing involves important safety principles and must meet regulatory and consumer standards. • Food processing has benefits (e.g. improved food safety) and risks (e.g. contamination, biological risks, environmental impact) that need careful management. **Ingredients and techniques** • Processing techniques (e.g. fermentation, dehydration, emulsification) alter ingredient properties to achieve specific outcomes. • Ingredient performance properties and processing techniques connect in the production of fit-for-purpose processed food outcomes. • Processing technologies are adopted for reasons that include yield, consistency, shelf life, labour cost, and energy use. These technologies may have other impacts to food outcomes, requiring observation and management. • Relevant feedback from stakeholders and users is combined with functional modelling results to determine refinements and evaluate overall fitness for purpose. **Responsible design** • Sustainable design of food outcomes must balance fitness for purpose with societal (e.g. user needs, cultural factors) and environmental (e.g. sustainable practices) considerations, and ethics to make decisions about processed food designs.	• Design and make an authentic, fit-for-purpose Food and Processing Technology outcome. **Codes of practice** • Model safe working practices to manage health risks associated with food processing (e.g. cross-contamination, safe storage and processing of food, knowledge of the danger zone). • Follow codes of practice to safely use processing equipment to produce fit-for-purpose outcomes. **Ingredients and techniques** • Analyse existing solutions to inform ingredient selection, considering the specifications and constraints of a given brief. • Experiment with ingredients to discover their performance properties suitable for fit-for-purpose outcomes. • Select and use ingredients based on how performance properties work together to achieve desired attributes and specifications through processing methods. • Conduct functional modelling trials to compare how processing technique and ingredient options contribute to desired outcomes. **Responsible design** • Record observations about responsible commercial decisions and suggest simple improvements to reduce waste or environmental impact.	• Design and make an authentic, fit-for-purpose Food and Processing Technology outcome. **Codes of practice** • Follow school policies and procedures to ensure safe production of fit-for-purpose outcomes. • Identify and manage food processing risks by using equipment and managing ingredients safely and appropriately, minimising risks to people and property. **Ingredients and techniques** • Trial how processing techniques alter ingredient performance properties in relation to specifications when developing fit-for-purpose outcomes. • Analyse functional modelling results comparing ingredients and techniques trialled in the development of fit-for-purpose food outcomes. • Implement processing technologies and analyse the effect on efficiency, quality, and sustainability when producing a fit-for-purpose outcome. • Analyse feedback and results from functional modelling to make informed decisions for refinement and make a judgement of overall fitness for purpose. **Responsible design** • Justify food outcome design decisions using user needs, stakeholder feedback, functional modelling evidence, and processing knowledge.

Systems and Control

	Knowledge *The facts, concepts, principles, and theories to teach*		Practices *The skills, strategies, and applications to teach*	
	During Year 9	**During Year 10**	**During Year 9**	**During Year 10**
	The content within this strand is underpinned by the common body of knowledge and practice in the Design, Make, and Innovate strand. These foundational design and development processes are to be applied here in the context of creating authentic and fit-for-purpose Systems and Control outcomes. Safe and purposeful practice includes identifying, assessing, and managing risks, and applying relevant health and safety procedures when designing, constructing, testing, troubleshooting, and evaluating mechanical, electronic, and mechanical systems. Knowledge and practices related to programmed components can be taught alongside Digital Technologies, which covers programming. This content can also be taught independently using pre-programmed components.			
	Systems thinking • Systems are designed to transform inputs into outputs that meet an identified purpose (e.g. automated gate, irrigation system, security system). • Mechanical, structural, electronic, and digital subsystems, including programmable control elements, can be integrated into mechatronic systems that perform coordinated tasks. • Feedback from within and outside of the system provides information about performance and supports monitoring, maintenance, adjustment, and improvement. • System representations (e.g. system diagrams, circuit diagrams, annotated sketches) communicate relationships between components, subsystems, and flows of energy or information (see Year 9 Science: Electric Charge and Energy Transfer in Circuits). **Physical systems** • Electrical and electronic systems require suitable power sources, components, and connections to operate safely and reliably. • Mechanical performance is influenced by forces, movement, friction, stability, alignment, and material selection. • Testing identifies faults and performance issues that can be used to improve system function and reliability (e.g. misalignment, loose connections, excessive friction, incorrect wiring).	**Systems thinking** • Integrated mechatronic systems combine mechanical, structural, electronic, and digital components that work together to achieve a purpose. • Feedback loops enable systems to respond to changing conditions and influence reliability, safety, efficiency, maintenance, and user experience. • Microcontrollers and embedded control systems use programmed instructions to receive input from sensors, process information, make decisions, and control outputs. • System performance can be evaluated using testing, user feedback, and performance data. **Physical systems** • Sensors detect environmental or user inputs that can be used to control system behaviour (e.g. light, temperature, distance, moisture, movement). • Actuators convert electrical energy into mechanical movement. Other devices convert it into outputs (e.g.light). • Malfunctions and failures may occur within mechanical, structural, electrical, electronic, or control subsystems and affect overall system performance and reliability.	• Design and make an authentic, fit-for-purpose Systems and Control outcome. **Systems thinking** • Analyse an existing system to explain how components and subsystems interact to achieve a purpose. • Use system diagrams, circuit diagrams, annotated sketches, and sequenced instructions to communicate how a system operates. • Evaluate system performance against identified functional, user, safety, and environmental requirements. • Record and interpret testing evidence to explain system behaviour and inform improvements. **Physical systems** • Select appropriate components, mechanisms, structures, and materials for an intended system outcome. • Identify, diagnose, and correct faults in simple electrical, mechanical, or structural systems. • Refine system performance by addressing issues relating to reliability, stability, friction, alignment, movement, or component connections.	• Design and make an authentic, fit-for-purpose Systems and Control outcome. **Systems thinking** • Design and communicate integrated mechatronic systems using system diagrams, annotated sketches, models, digital representations, or computer-aided design (CAD). • Develop systems that use sensors and feedback to respond to changing conditions. • Collect, interpret, and use testing evidence or simple performance data to evaluate how effectively a system responds to changing conditions. • Refine and justify system improvements using evidence from testing and stakeholder feedback. **Physical systems** • Use programmed microcontrollers or embedded control systems to control outputs. • Integrate sensors, actuators, and programmed control components into a functioning mechatronic system that responds to changing conditions (e.g. automated lighting, environmental monitoring, automated access systems). • Troubleshoot malfunction and failures in mechanical, electrical, electronic, structural, or control subsystems.

Digital Technology

	Knowledge *The facts, concepts, principles, and theories to teach*		Practices *The skills, strategies, and applications to teach*	
	During Year 9	**During Year 10**	**During Year 9**	**During Year 10**
	The content within this strand is underpinned by the common body of knowledge and practice in the Design, Make, and Innovate strand. These foundational design and development processes are to be applied here in the context of creating authentic and fit-for-purpose Digital Technology outcomes. Responsible practice includes considering accessibility, privacy, consent, ownership, attribution, cultural perspectives, appropriate use of AI and emerging technologies, and the potential impacts of digital outcomes on individuals, communities, and society.			
	Computer science • Computational thinking uses decomposition, abstraction, pattern recognition, algorithms, and evaluation to analyse problems for digital solutions. • Digital systems represent data using types and formats, and these choices affect how information is stored, processed, displayed, and used. **Programming** • Programs use sequence, selection, iteration, variables, input/output, and reusable structures to solve problems for a purpose, user, and context. • Program quality is identified by clear structure, readable naming, comments, testing, and debugging. **Digital media** • Digital media outcomes are shaped by purpose, audience, context, platform, conventions, design principles, and responsible practice. • Digital media development uses tools, assets, file formats, organised workflows, versions, exports, and evidence to support quality and continuity. **Responsible practice** • Digital outcomes can affect individuals, communities, and cultures in different ways depending on their design, purpose, and use.	**Computer science** • Algorithms and programs can be represented, traced, and compared for correctness, clarity, efficiency, and suitability for a specific problem. • Digital systems represent information using data, including binary, decimal, hexadecimal, text encoding, image or colour values. **Programming** • Programs are designed to be understandable, adaptable and maintainable through structure, functions, modularity, naming, comments, testing, and debugging. • Testing uses expected, boundary, and invalid inputs to check whether a program behaves as intended. **Digital media** • Digital media outcomes may be interactive, information-based, or time-based, and each type uses different conventions, design principles, and technical requirements. • A manageable digital project has a brief, purpose, user or audience, context, platform, requirements, scope, constraints, timeline, and responsible practice considerations. **Responsible practice** • AI and emerging technologies use data, models, prompts or automation, and raise questions about accuracy, bias, privacy, authorship, cultural representation, and who may benefit or be harmed.	• Design and make an authentic, fit-for-purpose Digital Technology outcome. **Computer science** • Explain simple problems by identifying purpose, users, inputs, outputs, constraints, context, and key steps. • Interpret and use simple data representations, such as text, numbers, Boolean values, images and colour values. **Programming** • Design, develop, test, and debug programs using sequence, selection, iteration, variables, input/output, and reusable structures. • Design, test, debug, and refine algorithms using flowcharts, pseudocode, and trace tables for tasks such as calculating totals or averages, validating user input, checking conditions, finding values in a list, or controlling a simple digital process. • Apply programming practices, including clear structure, meaningful naming, comments, testing evidence, and debugging. **Digital media** • Investigate existing digital media outcomes and apply relevant conventions, design principles, and technical skills to design and develop outcomes. • Create, manage, and refine digital assets and files using appropriate tools, formats, naming, versions, exports, backups, and evidence. **Responsible practice** • Apply appropriate processes when creating, sourcing, modifying, using, or sharing digital content, code, and data throughout the development of digital outcomes.	• Design and make an authentic, fit-for-purpose Digital Technology outcome. **Computer science** • Represent, trace, and compare algorithms using plain language, pseudocode, flowcharts, or code. • Convert, interpret, and explain selected data representations, and choose suitable data types for a programming or data-handling purpose. **Programming** • Develop programs using variables, selection, iteration, functions, input/output, data types, and simple modularity. • Test programs using expected, boundary, and invalid inputs, then use evidence to troubleshoot, refine, and explain changes to program logic or behaviour. **Digital media** • Investigate existing interactive, information-based, or time-based digital media outcomes, and apply relevant conventions, design principles, and technical skills to a selected context. • Design, develop, test, and refine a manageable digital outcome using a brief, requirements, scope, design representations, technical trialling, user or audience feedback, organised workflow, and evidence. **Responsible practice** • Evaluate digital, AI, or emerging technology outputs using trusted sources, human judgement, declared use, and responsible decision-making, including consideration of accuracy, bias, privacy, authorship, cultural representation, and potential benefits and harms.

Understanding learning progress

This content is currently being developed.